

## INK HEATER SYSTEM BALANCED

This document is to understand how to heat ink with 6 heaters using a PSU lower than total power required.

Ink Heater system balanced is implemented on P4 code. It is based on resistors and temperature sensors. Same Power Supply is used in 8 print bars and for this reason, is necessary to limit our max power supplied applied to each bar

Next table is to start explaining how a power factor balanced for reducing Duty Cycle is calculated. Check total power will be always lower than max available

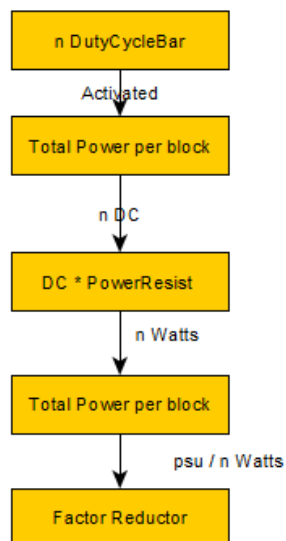
| DC (%) |           |     | PowerConsumption |  | factor  |            | PowerConsumption |
|--------|-----------|-----|------------------|--|---------|------------|------------------|
| 0      | 0         | 0   | 0                |  | 0       | 1          | 0                |
| 10     | 0,1       | 0,8 | 640              |  | 4,6875  | 1          | 320              |
| 20     | 0,2       | 1,6 | 1280             |  | 2,34375 | 1          | 640              |
| 30     | 0,3       | 2,4 | 1920             |  | 1,5625  | 1          | 960              |
| 40     | 0,4       | 3,2 | 2560             |  | 1,17188 | 1          | 1280             |
| 50     | 0,5       | 4   | 3200             |  | 0,9375  | 0,9375     | 1600             |
| 60     | 0,6       | 4,8 | 3840             |  | 0,78125 | 0,78125    | 1920             |
| 70     | 0,7       | 5,6 | 4480             |  | 0,66964 | 0,66964286 | 2240             |
| 80     | 0,8       | 6,4 | 5120             |  | 0,58594 | 0,5859375  | 2560             |
| 90     | 0,9       | 7,2 | 5760             |  | 0,52083 | 0,52083333 | 2880             |
| 100    | 1         | 8   | 6400             |  | 0,46875 | 0,46875    | 3200             |
|        |           |     |                  |  |         |            |                  |
|        | Printbars |     | PowerResist      |  |         |            | PowerResist      |
|        | 8         |     | 800              |  |         |            | 400              |
|        |           |     |                  |  |         |            |                  |
|        | PsuPower  |     |                  |  |         |            |                  |
|        | 3000      |     |                  |  |         |            |                  |

It is allowing hardware optimization because using one PSU for all heaters

A Specific functional chart is explaining this functionality in Block Program

## BlockCtrl Prg

Power Limited with factor reductor implemented on DutyCycle



$\text{MIN} ( (\text{MaxPSU perBlock} / \text{TotalPowerPerBlock}), 1.0 )$



## Conclusions

This concept must be applied to 8 tanks in Schneider code and also P4, it means both platforms has to use same structure text FBs [ Implementation as, commandBool() ]

## Repository

```
FUNCTION timeToStartPWM
|
|// iItemNumber : it comes from mapping: This value come in block prg and it is from 0 to 5,
|//               because we have 6 Heater per block in P4
|// iTotalItems : it comes from block or printer, 6 bars per block in P4 example
|// iTaskCycle : defined by us in Sft CPU, it is 72ms and hardcoded to 149
|// iMsInCycle : current time
|
|MsInCycle := DINT_TO_UINT(TIME_TO_DINT(clock_ms()) MOD (REAL_TO_DINT(lc_rCyclePwm) * 1000) );
|
|timeToStartPWM := iMsInCycle >= iItemNumber * (lc_rCyclePwm * 1000 / iTotalItems) AND
|                 iMsInCycle <= (iItemNumber * (lc_rCyclePwm * 1000 / iTotalItems) + iTaskCycle);
|
END_FUNCTION
```

| Mapping | Map-1 |     | mod |            | B1 | B2 | B3 | B4 | B5 | B6 |
|---------|-------|-----|-----|------------|----|----|----|----|----|----|
| 0       | -1    |     | 0   | 0          |    |    |    |    |    |    |
| 1       | 0     |     | 1   | 333,33333  |    |    |    |    |    |    |
| 2       | 1     |     | 2   | 666,66667  |    |    |    |    |    |    |
| 3       | 2     |     | 3   | 1000       |    |    |    |    |    |    |
| 4       | 3     |     | 4   | 1333,33333 |    |    |    |    |    |    |
| 5       | 4     | BK1 | 5   | 1666,66667 |    |    |    |    |    |    |
| 6       | 5     |     | 0   | 2000       |    |    |    |    |    |    |
| 7       | 6     |     | 1   | 2333,33333 |    |    |    |    |    |    |
| 8       | 7     |     | 2   | 2666,66667 |    |    |    |    |    |    |
| 9       | 8     |     | 3   | 3000       |    |    |    |    |    |    |
| 10      | 9     |     | 4   | 3333,33333 |    |    |    |    |    |    |
| 11      | 10    | BK2 | 5   | 3666,66667 |    |    |    |    |    |    |

